LECTURE: Hyponatremia - Quick Overview

LEARNING OBJECTIVES:
1. Define hyponatremia and classify by osmolality
2. Identify the three volume states and their causes
3. Know first-line treatment for each type

DEFINITION: Serum Na+ < 135 mEq/L

CLASSIFICATION BY VOLUME STATUS:

HYPOVOLEMIC (low total body Na+, more water loss):
- Causes: vomiting, diarrhea, diuretics, Addisons disease
- Urine Na+ < 20 = extrarenal loss; Urine Na+ > 20 = renal loss
- Treatment: isotonic saline (0.9% NaCl)

EUVOLEMIC (normal Na+, excess water):
- Causes: SIADH, hypothyroidism, psychogenic polydipsia
- SIADH: Urine osm > 100, Urine Na+ > 40
- Treatment: fluid restriction; severe = hypertonic saline

HYPERVOLEMIC (excess Na+ and water, more water):
- Causes: CHF, cirrhosis, nephrotic syndrome
- Treatment: fluid restriction + furosemide

DANGER: Correct Na+ no faster than 6-8 mEq/L per 24 hours
- Too fast = Osmotic Demyelination Syndrome (ODS)
- Triad: dysarthria, dysphagia, flaccid paralysis

KEY DRUGS:
- Tolvaptan (V2 receptor antagonist) - euvolemic/hypervolemic SIADH
- Demeclocycline - blocks ADH at collecting duct

WARD PEARL: Beer potomania - low solute intake impairs ability
to excrete free water, causing hyponatremia despite normal ADH.